

# **Mutual Fund Analytics Capstone Project Bluestock Fintech**

## **Final Project Report**

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## **Project:**

Mutual Fund Analytics Platform

## **Technology Stack**

- Python
- Pandas
- NumPy
- SQLite
- Power BI
- Jupyter Notebook

## **Submission Date**

July 2026

## Executive Summary

The Mutual Fund Analytics Capstone Project focuses on building a complete end-to-end analytics platform for analyzing mutual fund performance, investor behavior, portfolio composition, and market trends.

The project begins with an automated ETL pipeline that collects, cleans, validates, and stores mutual fund datasets in an SQLite database. Exploratory Data Analysis (EDA) was performed to understand fund categories, investor demographics, AUM distribution, SIP trends, and portfolio allocation.

Financial performance metrics such as CAGR, Volatility, Sharpe Ratio, Beta, Alpha, Maximum Drawdown, Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), and Rolling Sharpe Ratio were implemented to evaluate mutual fund performance.

Advanced analytics including Investor Cohort Analysis, SIP Continuity Analysis, Sector Concentration using Herfindahl-Hirschman Index (HHI), and a Risk-Based Mutual Fund Recommendation System were developed to provide meaningful investment insights.

An interactive four-page Power BI dashboard was designed to visualize fund performance, investor analytics, portfolio diversification, and SIP market trends using dynamic slicers, drill-through functionality, and KPI cards.

The completed solution demonstrates practical skills in data engineering, financial analytics, business intelligence, and dashboard development using industry-standard tools.

### 3. Project Overview

The Mutual Fund Analytics Capstone Project was developed to build a complete data analytics solution for analyzing the performance of Indian mutual funds. The project integrates data engineering, financial analytics, database management, and business intelligence to provide actionable insights for investors and fund managers.

The solution processes multiple datasets related to mutual fund schemes, NAV history, investor transactions, SIP inflows, portfolio holdings, benchmark indices, and fund performance. An ETL pipeline was implemented to clean, transform, validate, and load the data into an SQLite database.

The project also includes financial performance calculations, risk analysis, investor behavior analysis, portfolio concentration analysis, and an interactive Power BI dashboard for visualizing key business metrics.

### 4. Project Objectives

The primary objectives of this project are:

- Build an automated ETL pipeline for mutual fund datasets.
- Store cleaned data in an SQLite database.
- Perform Exploratory Data Analysis (EDA) to understand fund performance and investor behavior.
- Calculate financial performance metrics such as CAGR, Volatility, Sharpe Ratio, Beta, Alpha, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR).
- Develop advanced analytics including Investor Cohort Analysis, SIP Continuity Analysis, Sector Concentration Analysis, and Mutual Fund Recommendation System.
- Design an interactive Power BI dashboard with multiple analytical pages and dynamic filters.
- Generate meaningful insights to support investment decision-making.

## **5. Dataset Description**

The project utilizes multiple datasets representing different aspects of the mutual fund ecosystem.

### **Main Datasets Used**

#### **Fund Master**

- Scheme details
- Fund house
- Category
- Fund manager
- Expense ratio
- Risk category

#### **NAV History**

- Daily Net Asset Value (NAV)
- Historical performance
- Return calculations

#### **Scheme Performance**

- Returns across multiple time periods
- Benchmark comparison
- Performance ranking

#### **Investor Transactions**

- Purchase and redemption transactions
- SIP investments
- Investor demographics

#### **Portfolio Holdings**

- Company holdings
- Sector allocation
- Portfolio weights

## **Benchmark Indices**

- Market index values
- Comparative performance analysis

## **Monthly SIP Inflows**

- SIP investment trends
- Monthly contribution analysis

## **AUM by Fund House**

- Assets Under Management
- Fund house comparison

These datasets were cleaned, validated, transformed, and integrated to build a unified analytics platform for further analysis and dashboard development.

# **6. ETL Pipeline**

The ETL (Extract, Transform, Load) pipeline was developed to automate the preparation of mutual fund datasets for analysis.

## **Extract**

Data was collected from multiple CSV files containing information related to:

- Mutual Fund Master
- NAV History
- Scheme Performance
- Investor Transactions
- SIP Inflows
- Portfolio Holdings
- Benchmark Indices
- Assets Under Management (AUM)

## **Transform**

The following data cleaning operations were performed:

- Removal of duplicate records

- Handling missing values
- Standardization of date formats
- Conversion of numeric columns into appropriate data types
- Validation of fund identifiers
- Column name standardization
- Data consistency checks across datasets

## **Load**

The cleaned datasets were stored in an SQLite database for efficient querying and further analysis. This centralized database served as the primary data source for analytics notebooks and the Power BI dashboard.

The ETL pipeline was organized into modular Python scripts, allowing each dataset to be processed independently while maintaining consistency across the entire project.

## **7. Database Design**

SQLite was used as the database management system due to its lightweight architecture and ease of integration with Python.

The database stores all cleaned datasets in structured tables, enabling efficient retrieval and analytical processing.

### **Major Tables**

- Fund Master
- NAV History
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices
- SIP Inflows
- Category Inflows
- AUM by Fund House

Primary keys and common identifiers such as AMFI Code were used to establish relationships between datasets. The database design supports analytical queries, dashboard reporting, and performance metric calculations while minimizing data redundancy.

## **8. Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was performed to understand the characteristics of the mutual fund ecosystem before conducting advanced financial analysis.

The analysis focused on:

- Distribution of mutual fund categories
- Assets Under Management (AUM) across fund houses
- Historical NAV trends
- Monthly SIP inflows
- Investor demographic distribution
- Portfolio sector allocation
- Fund risk classification
- Benchmark performance comparison

Various visualizations such as bar charts, line charts, histograms, pie charts, scatter plots, and heatmaps were created to identify trends, outliers, and relationships within the data.

The EDA process provided valuable insights that guided subsequent performance analytics and dashboard development.

## **9. Performance Analytics**

The performance of mutual fund schemes was evaluated using multiple financial metrics to measure returns, volatility, and risk-adjusted performance.

The following metrics were implemented using Python:

### **Compound Annual Growth Rate (CAGR)**

CAGR was calculated to measure the annualized growth rate of mutual fund NAV over the investment period.

### **Daily Returns**

Daily percentage returns were computed from historical NAV values to support subsequent financial calculations.

### **Annualized Volatility**

Annualized volatility was calculated using the standard deviation of daily returns multiplied by the square root of 252 trading days.

### **Sharpe Ratio**

The Sharpe Ratio was used to evaluate risk-adjusted returns by comparing excess returns over the risk-free rate against annualized volatility.

### **Beta**

Beta measured the sensitivity of each mutual fund relative to its benchmark index.

### **Alpha**

Alpha represented the excess return generated by a fund compared to its expected return based on market movements.

### **Maximum Drawdown**

Maximum Drawdown measured the largest decline from a historical peak NAV, providing an estimate of downside risk.

These metrics enabled comprehensive evaluation of fund performance and were later integrated into the recommendation system and dashboard visualizations.

## **10. Advanced Analytics**

Several advanced analytical techniques were implemented to provide deeper investment insights.

### **Historical Value at Risk (VaR)**

Historical VaR at the 95% confidence level was calculated using historical daily returns to estimate the maximum expected loss under normal market conditions.

### **Conditional Value at Risk (CVaR)**

CVaR was computed to estimate the average loss beyond the Value at Risk threshold, providing a better understanding of tail risk.

### **Rolling 90-Day Sharpe Ratio**



Rolling Sharpe Ratio was calculated using a 90-day moving window to evaluate changes in risk-adjusted performance over time.

### **Investor Cohort Analysis**

Investor transactions were grouped based on investment year to analyze long-term investment behavior and average SIP contributions.

### **SIP Continuity Analysis**

Transaction history was analyzed to classify investors as Regular or At Risk based on average gaps between SIP investments.

### **Sector Concentration Analysis**

Portfolio holdings were analyzed using the Herfindahl-Hirschman Index (HHI) to measure sector diversification and concentration.

### **Mutual Fund Recommendation System**

A rule-based recommendation engine was developed to recommend suitable mutual funds based on investor risk appetite. Funds were ranked using calculated performance metrics and risk categories.

## **11. Dashboard Overview**

A four-page interactive Power BI dashboard was developed to visualize key business metrics and financial insights.

### **Page 1 – Fund Performance**

- Fund performance KPIs
- NAV trends
- Benchmark comparison
- Risk-adjusted performance indicators

### **Page 2 – Portfolio Analysis**

- Portfolio diversification
- Sector allocation
- Fund category distribution
- Holdings analysis

### **Page 3 – Investor Analytics**

- Investor demographics
- SIP amount analysis
- Cohort analysis
- Monthly transaction trends

#### **Page 4 – SIP & Market Trends**

- Monthly SIP inflows
- Benchmark index performance
- Market trend comparison
- Interactive slicers and drill-through functionality

The dashboard enables users to interactively explore fund performance, investor behavior, and market trends through dynamic visualizations and filters.

## **12. Key Insights**

The project generated several meaningful insights from the mutual fund datasets.

- Equity mutual funds generally demonstrated higher long-term returns but also exhibited greater volatility.
- Investors belonging to higher income groups contributed larger SIP amounts on average.
- Several investors showed irregular SIP contribution patterns, indicating potential churn risk.
- Sector concentration analysis revealed that some mutual funds were highly concentrated in Banking and Information Technology sectors, increasing portfolio risk.
- Funds with higher Sharpe Ratios consistently delivered better risk-adjusted returns.
- Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) highlighted the downside risk associated with different mutual fund categories.
- The recommendation system successfully identified suitable funds based on investor risk appetite and overall performance metrics.

## 13.Challenges Faced

During the development of the project, several challenges were encountered.

- Managing multiple datasets with different structures and formats.
- Handling missing values and inconsistent data across various data sources.
- Standardizing date formats and ensuring proper alignment between NAV and benchmark datasets.
- Calculating financial metrics accurately while handling missing trading days.
- Establishing relationships between datasets for dashboard reporting.
- Designing an interactive Power BI dashboard with meaningful visualizations and slicers.
- Organizing the project into a professional folder structure suitable for final submission.

These challenges were addressed through systematic data cleaning, validation, modular Python scripts, and iterative testing.

## 14. Conclusion

The Mutual Fund Analytics Capstone Project successfully demonstrates an end-to-end data analytics workflow covering data engineering, financial analysis, database management, and business intelligence.

The developed solution automates data preparation through an ETL pipeline, stores cleaned datasets in an SQLite database, performs comprehensive exploratory and financial analysis, and presents insights through an interactive Power BI dashboard.

Advanced analytical techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Investor Cohort Analysis, SIP Continuity Analysis, Sector Concentration Analysis, and Mutual Fund Recommendation further enhance the practical usefulness of the platform.

Overall, the project provides valuable investment insights and demonstrates strong proficiency in Python, SQL, Power BI, and data analytics techniques.